Lab6 Gene Expression

Dylan Renard

Professor Brandon Higgs

Gene Expression and Data Visualization

Johns Hopkins University

October 23rd, 2025

Lab6

In this lab, we will be working with an Affymetrix data set that was run on the human HGU95A array. This experiment was designed to assess the gene expression events in the frontal cortex due to aging. A total of 18 male and 12 female postmortem brain samples were obtained to assess this.

The analysis that we are interested in conducting is a direct follow up to the previous lab of differential expression. We first want to identify those genes/probes that are differentially expressed in the frontal cortex between old and young subjects, then between males and females. Next, we would like to evaluate the differences between a couple of multiple testing adjustment methods. As explained in the lecture and the course website, multiple testing is a necessary step to reduce false positives when conducting more than a single statistical test. You will generate some p-value plots to get an idea of the how conservative some methods are compared to others.

I have identified 2 gene vectors for you to use below, so do not calculate the t-test or adjustments on the entire array of genes/probes.

For the second part of this lab, you will be working with RNA-sequencing data from The Cancer Genome Atlas (TCGA), specifically a breast invasive carcinoma dataset of 119 patient tumors. The data matrix and annotation files are on the course website. We will be trying to confirm an observation from a meta-analysis performed by Mehra et al, 2005 in Cancer Research. The authors identified the gene (using arrays) and protein (using immunohistochemistry) GATA3 as a prognostic factor in breast cancer, where patients with low expression of GATA3 experienced overall worse survival. The PubMed abstract is here: http://www.ncbi.nlm.nih.gov/pubmed/16357129.

Q1) Download the GEO Brain Aging study from the class website. Also obtain the annotation file for this data frame.

Q2) Load into R, using read.table() function and the

header=T/row.names=1 arguments for each data file.

▼ Code

```
## Q1–Q2
library(DT)


# Gender-associated probes
g.g <- c(1394,1474,1917,2099,2367,2428,2625,3168,3181,3641,3832,4526,4731,
         4863,6062,6356,6684,6787,6900,7223,7244,7299,8086,8652,8959,9073,
         9145,9389,10219,11238,11669,11674,11793)

# Age-associated probes
g.a <- c(25,302,1847,2324,246,2757,3222,3675,4429,4430,4912,5640,5835,
         5856,6803,7229,7833,8133,8579,8822,8994,10101,11433,12039,12353,
         12404,12442,67,88,100)

affy_expr <- read.table("agingStudy11FCortexAffy.txt",
                        header = TRUE, row.names = 1, sep = "\t")

affy_anno <- read.table("agingStudy1FCortexAffyAnn.txt",
                        header = TRUE, sep = "\t")



dim(affy_expr)
```

```
[1] 12625    30
```

▼ Code

```
head(affy_expr[, 1:5])
```

```
           GSM27015.26.M GSM27016.26.M GSM27018.29.M GSM27021.37.M
31307_at         179.8630      106.4950      265.5860      301.2430
31308_at         559.0780      411.4830      481.1760      570.7330
31309_r_at        20.7697       30.6415       50.2153       42.6892
31310_at         154.1910      224.4460      188.8230      177.8630
31311_at         956.7970      648.3100      933.6560     1016.4100
31312_at         186.5800      150.0220      262.3690      203.9770
           GSM27023.40.M
31307_at         218.5090
31308_at         333.5390
31309_r_at        27.1059
31310_at         233.4630
31311_at         762.0130
31312_at         169.4220
```

▼ Code

```
head(affy_anno)
```

```
        ID Gender Age
1 GSM27015      M  26
2 GSM27016      M  26
3 GSM27018      M  29
4 GSM27021      M  37
5 GSM27023      M  40
6 GSM27024      M  42
```

Q3) Prepare 2 separate vectors for comparison. The first is a comparison between male and female patients. The current data frame can be left alone for this, since the males and females are all grouped together. The second vector is comparison between patients >= 50 years of age and those < 50 years of age. To do this, you must use the annotation file and logical operators to isolate the correct arrays/samples.

▼ Code

```
## Q3) Comparison vectors (M vs F) and (>=50 vs <50)

# 1) Extract GSM IDs from expression column names (drop ".age.sex")
expr_ids_full <- colnames(affy_expr)
expr_ids_gsm  <- sub("\\..*$", "", expr_ids_full)  # "GSM27015" from "GSM27015.26.M"

# 2) Align annotation to expression columns by GSM ID
affy_anno <- affy_anno[match(expr_ids_gsm, affy_anno$ID), ]
stopifnot(identical(affy_anno$ID, expr_ids_gsm))

# 3) Build groups
sex_fac <- factor(affy_anno$Gender, levels = c("M","F"))
age_num <- as.numeric(affy_anno$Age)
age_fac <- factor(ifelse(age_num >= 50, ">=50", "<50"), levels = c("<50", ">=50"))

idx_male   <- which(sex_fac == "M")
idx_female <- which(sex_fac == "F")
idx_lt50   <- which(age_fac == "<50")
idx_ge50   <- which(age_fac == ">=50")

wrap_html <- function(v) sprintf("<div style='white-space:normal;'>%s</div>", paste(v, co

# --- Sex table ---
sex_tbl <- data.frame(
```

```
  Group   = c("M", "F"),
  Qty     = c(length(idx_male), length(idx_female)),
  Samples = c(wrap_html(expr_ids_full[idx_male]),
              wrap_html(expr_ids_full[idx_female])),
  stringsAsFactors = FALSE
)

# --- Age table ---
age_tbl <- data.frame(
  Group   = c("<50", ">=50"),
  Qty     = c(length(idx_lt50), length(idx_ge50)),
  Samples = c(wrap_html(expr_ids_full[idx_lt50]),
              wrap_html(expr_ids_full[idx_ge50])),
  stringsAsFactors = FALSE
)

# --- Render clean tables ---
datatable(
  sex_tbl, rownames = FALSE, escape = FALSE,
  options = list(paging = FALSE, dom = 't', scrollX = TRUE)
)
```

Group	Qty	Samples
M	18	GSM27015.26.M, GSM27016.26.M, GSM27018.29.M, GSM27021.37.M, GSM27023.40.M, GSM27024.42.M, GSM27025.45.M, GSM27027.52.M, GSM27028.53.M, GSM27031.66.M, GSM27032.70.M, GSM27034.73.M, GSM27035.77.M, GSM27036.80.M, GSM27038.85.M, GSM27040.90.M, GSM27042.91.M, GSM27043.95.M
F	12	GSM27017.27.F, GSM27019.30.F, GSM27020.36.F, GSM27022.38.F, GSM27026.48.F, GSM27029.56.F, GSM27030.61.F, GSM27033.71.F, GSM27037.81.F, GSM27039.87.F, GSM27041.90.F, GSM27044.106.F

▼ Code

```
datatable(
  age_tbl, rownames = FALSE, escape = FALSE,
  options = list(paging = FALSE, dom = 't', scrollX = TRUE)
)
```

Group	Qty	Samples
<50	12	GSM27015.26.M, GSM27016.26.M, GSM27018.29.M, GSM27021.37.M, GSM27023.40.M, GSM27024.42.M, GSM27025.45.M, GSM27017.27.F, GSM27019.30.F, GSM27020.36.F, GSM27022.38.F, GSM27026.48.F

>=50	18	GSM27027.52.M, GSM27028.53.M, GSM27031.66.M, GSM27032.70.M, GSM27034.73.M, GSM27035.77.M, GSM27036.80.M, GSM27038.85.M, GSM27040.90.M, GSM27042.91.M, GSM27043.95.M, GSM27029.56.F, GSM27030.61.F, GSM27033.71.F, GSM27037.81.F, GSM27039.87.F, GSM27041.90.F, GSM27044.106.F

Q4) Run the t.test function from the notes using the first gene vector below for the gender comparison. Then use the second gene vector below for the age comparison. Using these p-values, use either p.adjust in the base library or mt.rawp2adjp in the multtest library to adjust the values for multiple corrections with the Holm's method.

▼ Code

```
## Q4) t-tests and Holm's correction
# Function to run unpaired equal-variance t-tests for a set of probes
run_ttests <- function(expr, probe_idx, group_factor) {
  sapply(probe_idx, function(i) {
    x <- as.numeric(expr[i, ])
    t.test(x ~ group_factor, var.equal = TRUE)$p.value
  })
}

# --- (a) Gender comparison ---
p_gender <- run_ttests(affy_expr, g.g, sex_fac)
p_gender_holm <- p.adjust(p_gender, method = "holm")

# Combine into a results data frame
res_gender <- data.frame(
  Probe = rownames(affy_expr)[g.g],
  Raw_p  = signif(p_gender, 4),
  Holm_p = signif(p_gender_holm, 4)
)

# --- (b) Age comparison ---
p_age <- run_ttests(affy_expr, g.a, age_fac)
p_age_holm <- p.adjust(p_age, method = "holm")

res_age <- data.frame(
  Probe = rownames(affy_expr)[g.a],
  Raw_p  = signif(p_age, 4),
  Holm_p = signif(p_age_holm, 4)
)
datatable(
  res_gender, rownames = FALSE,
  caption = "Q4a. Gender comparison: raw and Holm-adjusted p-values",
```

```
  options = list(paging = FALSE, dom = 't')
)
```

Probe	Raw_p	Holm_p
35570_at	0.009133	0.09613
36367_at	0.00007329	0.001832
33937_at	0.007743	0.09613
34477_at	1.173e-10	3.753e-9
35465_at	0.00143	0.03289
35885_at	4.843e-9	1.453e-7
36321_at	6.369e-7	0.00001656
38182_at	0.004203	0.07565
38195_at	0.009129	0.09613
39613_at	0.003275	0.07205
40044_at	0.007114	0.09613
31859_at	0.003666	0.07332
32660_at	0.004344	0.07565
33269_at	0.007389	0.09613
37583_at	1.093e-8	3.059e-7
38355_at	8.558e-9	2.482e-7
39402_at	0.006719	0.09613
39744_at	0.006297	0.09613
40097_at	4.061e-8	0.000001096
41138_at	0.001175	0.02819
41159_at	0.003831	0.07332
41214_at	3.498e-17	1.154e-15

Probe	Raw_p	Holm_p
34870_at	0.008625	0.09613
36995_at	0.009904	0.09613
38020_at	0.007359	0.09613
38374_at	0.006731	0.09613
38446_at	1.466e-9	4.545e-8
39168_at	0.006008	0.09613
41556_s_at	0.006263	0.09613
1425_at	0.00612	0.09613
963_at	0.009966	0.09613
968_i_at	0.003281	0.07205
865_at	0.009987	0.09613

Q4a. Gender comparison: raw and Holm-adjusted p-values

▼ Code

```
datatable(
  res_age, rownames = FALSE,
  caption = "Q4b. Age comparison: raw and Holm-adjusted p-values",
  options = list(paging = FALSE, dom = 't')
)
```

Probe	Raw_p	Holm_p
31331_at	0.1889	0.9444
31608_g_at	0.00002324	0.0005346
33508_at	0.0000201	0.0004823
35422_at	0.00006329	0.0006961
31552_at	0.8325	1
37053_at	0.00002362	0.0005346

Probe	Raw_p	Holm_p
38236_at	0.0000112	0.0003249
39647_s_at	0.00004043	0.0006331
41719_i_at	0.00003957	0.0006331
41720_r_at	0.00001194	0.0003344
33318_at	0.00002993	0.0005986
36083_at	0.00003314	0.0005986
36876_at	0.00008613	0.0006961
36897_at	0.00002402	0.0005346
39760_at	0.00009338	0.0006961
41144_g_at	0.00007431	0.0006961
33899_at	0.00005165	0.0006715
35277_at	0.00001446	0.0003904
36681_at	0.000008052	0.0002416
37405_at	0.00005475	0.0006715
38055_at	0.00004248	0.0006331
41318_g_at	0.0000734	0.0006961
1217_g_at	0.00001576	0.0004012
622_at	0.00003639	0.0006186
275_at	0.00001543	0.0004012
235_at	0.00003101	0.0005986
183_at	0.00006586	0.0006961
31373_at	0.8317	1
31394_at	0.9163	1
31406_at	0.5755	1

Q4b. Age comparison: raw and Holm-adjusted p-values

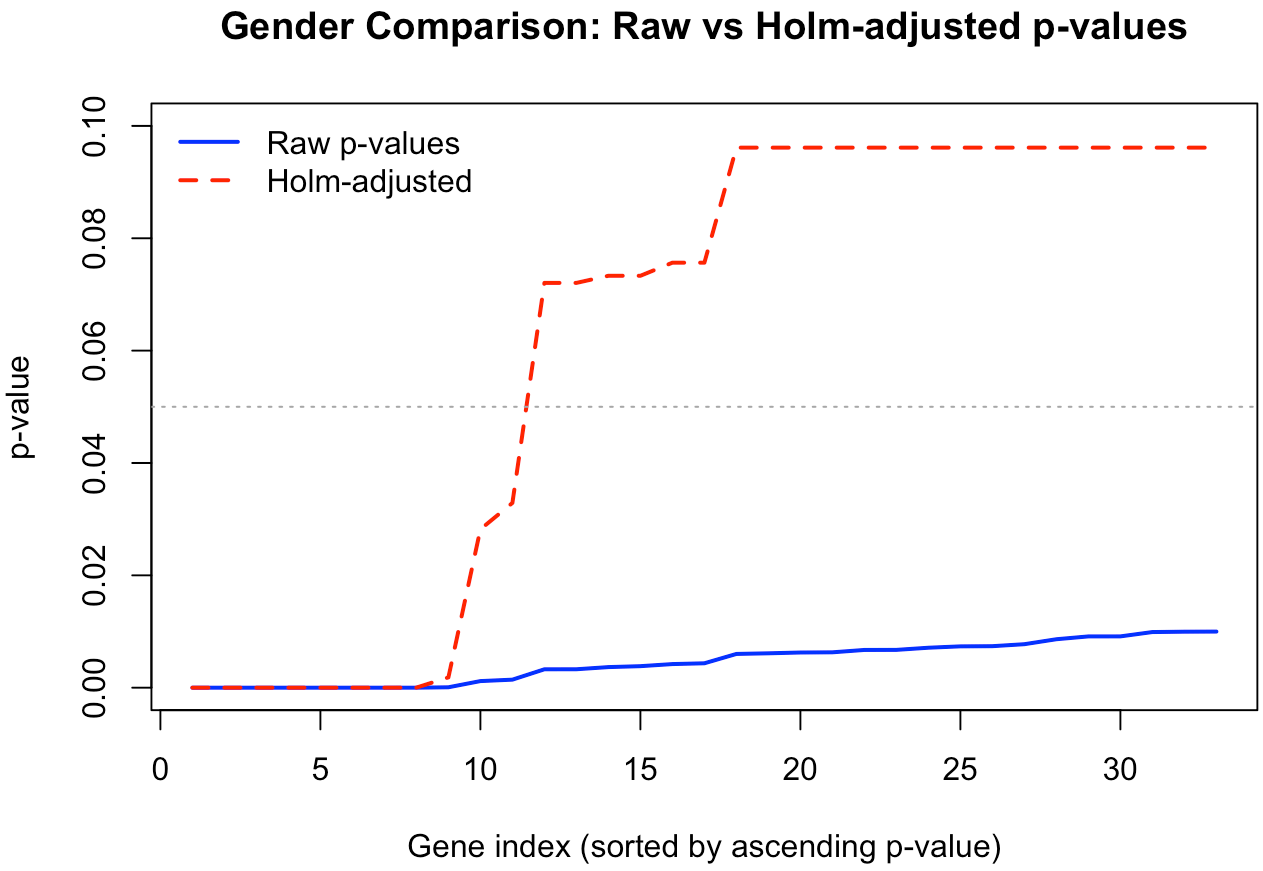
Q5) Sort the adjusted p-values and non-adjusted p-values and plot them vs. the x-axis of numbers for each comparison data set. Make sure that the two lines are different colors. Also make sure that the p-values are sorted before plotting.

▼ Code

```
## Q5) Sorted p-values: raw vs Holm-adjusted

# --- Gender comparison ---
ord_gender_raw  <- sort(p_gender)
ord_gender_holm <- sort(p_gender_holm)
n_gender <- length(ord_gender_raw)

plot(1:n_gender, ord_gender_raw, type = "l", col = "blue", lwd = 2,
     xlab = "Gene index (sorted by ascending p-value)",
     ylab = "p-value",
     ylim = c(0,0.1),
     main = "Gender Comparison: Raw vs Holm-adjusted p-values")
lines(1:n_gender, ord_gender_holm, col = "red", lwd = 2, lty = 2)
abline(h = 0.05, col = "darkgray", lty = 3)
legend("topleft", legend = c("Raw p-values", "Holm-adjusted"),
       col = c("blue", "red"), lwd = 2, lty = c(1,2), bty = "n")
```



▼ Code

```
# --- Age comparison ---
ord_age_raw  <- sort(p_age)
ord_age_holm <- sort(p_age_holm)
n_age <- length(ord_age_raw)

plot(1:n_age, ord_age_raw, type = "l", col = "blue", lwd = 2,
     xlab = "Gene index (sorted by ascending p-value)",
     ylab = "p-value",
     ylim = c(0,0.1),
     main = "Age Comparison: Raw vs Holm-adjusted p-values")
lines(1:n_age, ord_age_holm, col = "red", lwd = 2, lty = 2)
abline(h = 0.05, col = "darkgray", lty = 3)
legend("topleft", legend = c("Raw p-values", "Holm-adjusted"),
       col = c("blue", "red"), lwd = 2, lty = c(1,2), bty = "n")
```

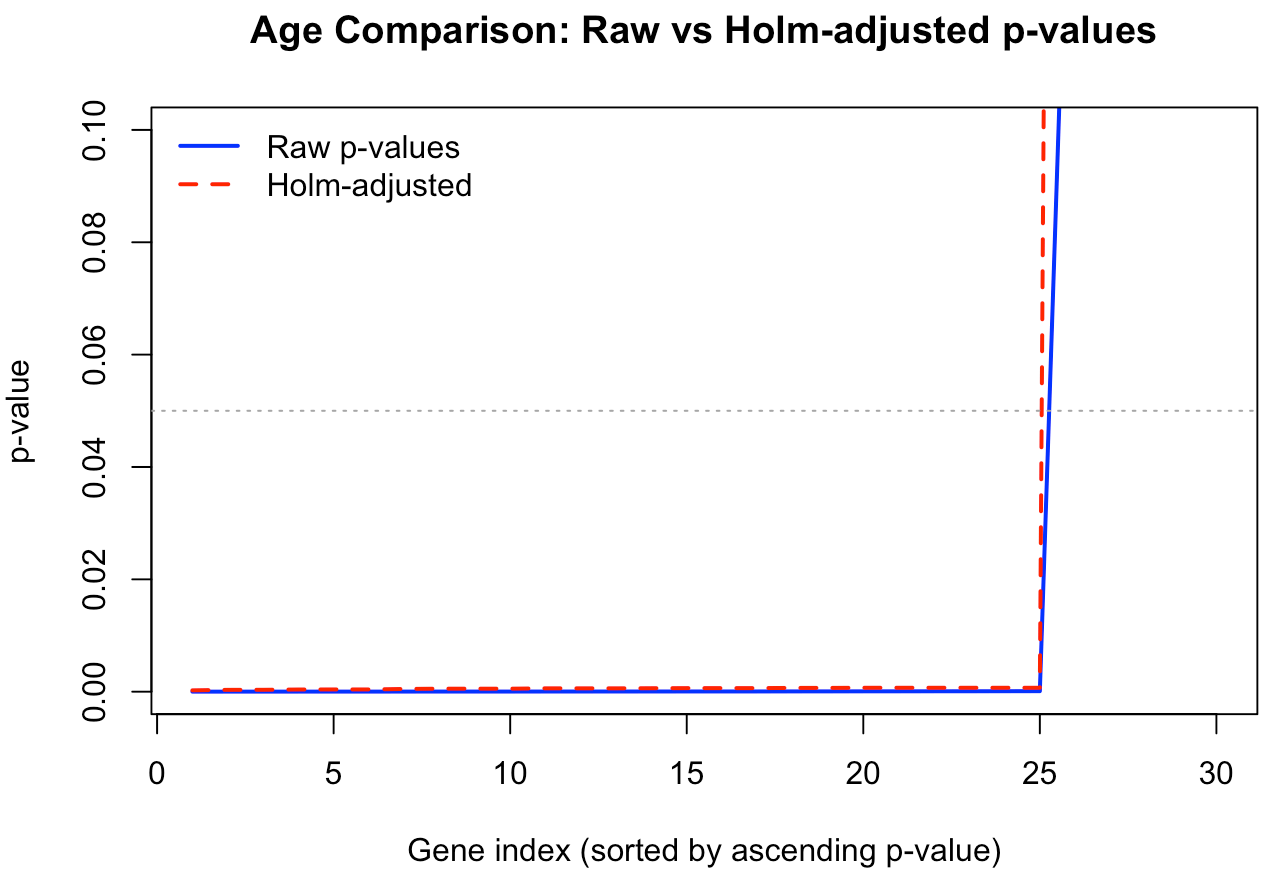


Figure Q5 Interpretation: Sorted p-value plots for the gender and age comparisons, each restricted to a 0–0.05 y-axis range to highlight the significance region. The x-axis represents the gene index ranked by ascending p-value (1 = smallest, N = largest). Blue lines show raw p-values, and red dashed lines show Holm-adjusted p-values. In the gender dataset, several genes approach or fall below the 0.05 threshold before correction, suggesting potential sex-associated expression differences. After Holm adjustment, these p-values increase but a few may remain near the cutoff, indicating marginal significance. The age comparison shows generally higher p-values, with both raw and adjusted values clustering near zero for only a small subset of probes. Overall, the Holm correction reduces apparent significance and better reflects the limited sample size and multiple-testing context emphasized in the lecture examples.

Q6) Repeat #4 and #5 with the Bonferroni method.

▼ Code

```
## Q6) Repeat analysis using Bonferroni correction (y-axis 0–0.10)

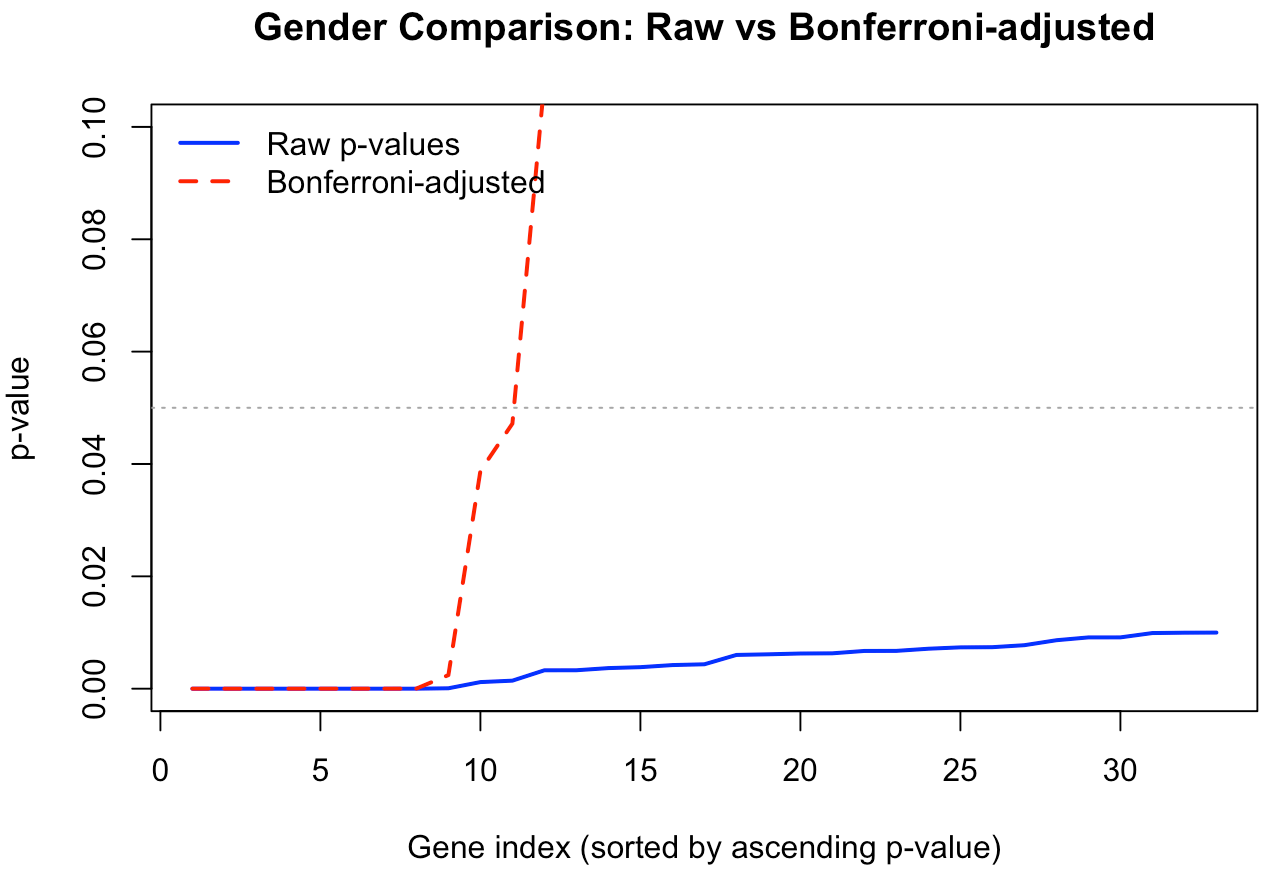
# --- (a) Gender comparison ---
p_gender_bonf <- p.adjust(p_gender, method = "bonferroni")

# --- (b) Age comparison ---
p_age_bonf <- p.adjust(p_age, method = "bonferroni")
```

```
# --- Plot: Gender and Age (Raw vs Bonferroni) ---

# Gender
ord_gender_raw  <- sort(p_gender)
ord_gender_bonf <- sort(p_gender_bonf)
n_gender <- length(ord_gender_raw)

plot(1:n_gender, ord_gender_raw, type = "l", col = "blue", lwd = 2,
     ylim = c(0, 0.10),
     xlab = "Gene index (sorted by ascending p-value)",
     ylab = "p-value",
     main = "Gender Comparison: Raw vs Bonferroni-adjusted")
lines(1:n_gender, ord_gender_bonf, col = "red", lwd = 2, lty = 2)
abline(h = 0.05, col = "darkgray", lty = 3)
legend("topleft", legend = c("Raw p-values", "Bonferroni-adjusted"),
       col = c("blue", "red"), lwd = 2, lty = c(1,2), bty = "n")
```



▼ Code

```
# Age
ord_age_raw  <- sort(p_age)
ord_age_bonf <- sort(p_age_bonf)
n_age <- length(ord_age_raw)
```

```
plot(1:n_age, ord_age_raw, type = "l", col = "blue", lwd = 2,
     ylim = c(0, 0.10),
     xlab = "Gene index (sorted by ascending p-value)",
     ylab = "p-value",
     main = "Age Comparison: Raw vs Bonferroni-adjusted")
lines(1:n_age, ord_age_bonf, col = "red", lwd = 2, lty = 2)
abline(h = 0.05, col = "darkgray", lty = 3)
legend("topleft", legend = c("Raw p-values", "Bonferroni-adjusted"),
       col = c("blue", "red"), lwd = 2, lty = c(1,2), bty = "n")
```

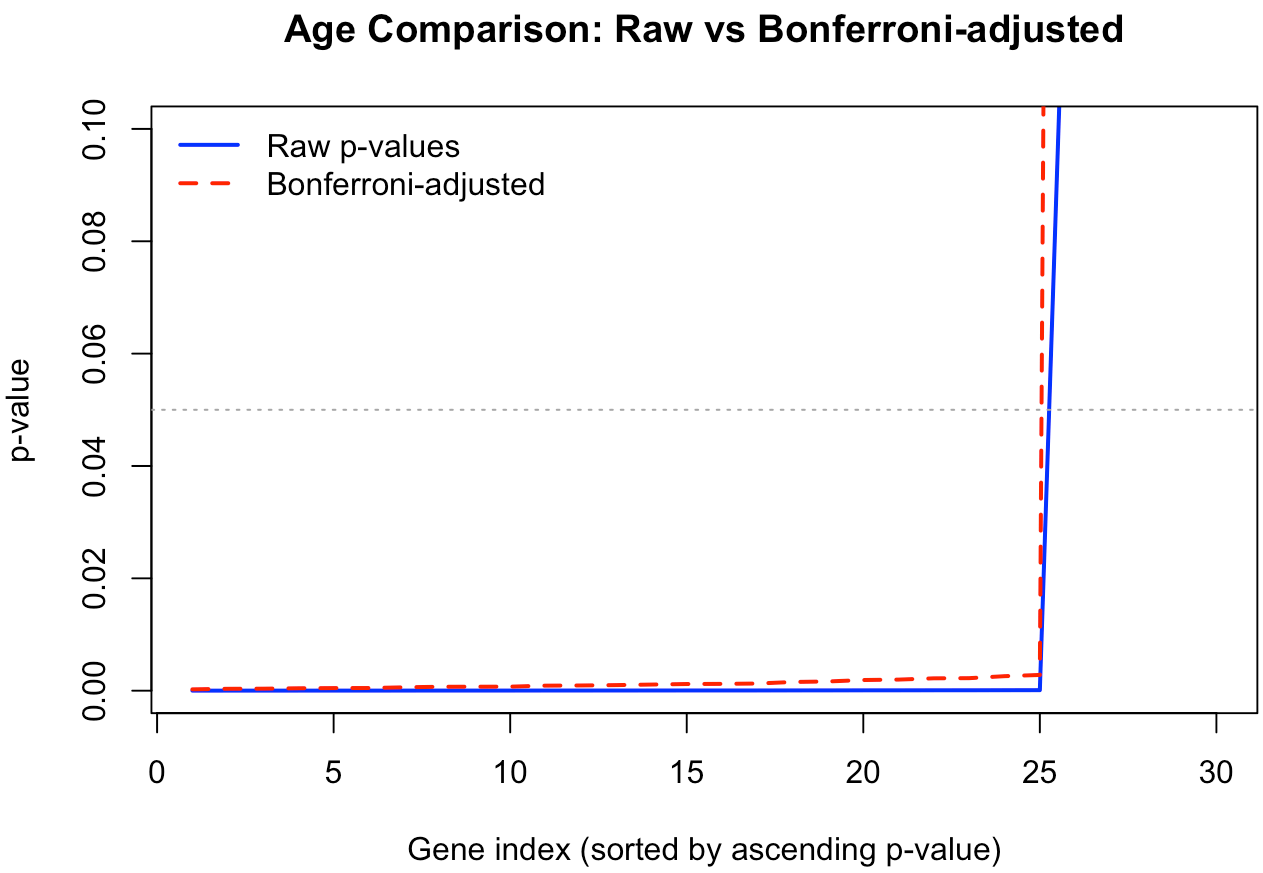


Figure 6 Interpretation: Figure 6 applies the Bonferroni correction to the same gender and age comparisons shown in Figure 5, replacing Holm's stepwise adjustment with Bonferroni's uniform scaling. As expected, Bonferroni inflates p-values more strongly, resulting in a steeper upward curve across the sorted gene index. In the gender comparison, the number of probes remaining below the 0.05 significance threshold decreased from twelve under Holm correction to nine under Bonferroni, reflecting the stricter nature of Bonferroni's family-wise error control. This inflation was most apparent among the mid- to higher-ranked genes, where several probes that were marginally significant under Holm were pushed above the cutoff after Bonferroni adjustment. In contrast, the age comparison remained stable across correction methods: the same small subset of probes—about five—retained high p-values, and the overall pattern of raw and adjusted distributions changed little. Together, these results demonstrate that Bonferroni correction more aggressively reduces apparent significance in the gender dataset, while

datasets with fewer borderline findings, such as the age comparison, remain largely unaffected by the choice of correction method.

▼ Code

```
# Threshold for significance
alpha <- 0.05

# --- Gender comparison ---
sig_gender_idx <- which(p_gender_bonf < alpha)
sig_gender_tbl <- data.frame(
  Probe_ID = rownames(affy_expr)[g.g[sig_gender_idx]],
  Raw_p    = p_gender[sig_gender_idx],
  Bonf_p   = p_gender_bonf[sig_gender_idx],
  stringsAsFactors = FALSE
)

# Sort from smallest to largest adjusted p-value
sig_gender_tbl <- sig_gender_tbl[order(sig_gender_tbl$Bonf_p), ]
sig_gender_tbl$Raw_p  <- signif(sig_gender_tbl$Raw_p, 4)
sig_gender_tbl$Bonf_p <- signif(sig_gender_tbl$Bonf_p, 4)

# --- Age comparison ---
sig_age_idx <- which(p_age_bonf < alpha)
sig_age_tbl <- data.frame(
  Probe_ID = rownames(affy_expr)[g.a[sig_age_idx]],
  Raw_p    = p_age[sig_age_idx],
  Bonf_p   = p_age_bonf[sig_age_idx],
  stringsAsFactors = FALSE
)

sig_age_tbl <- sig_age_tbl[order(sig_age_tbl$Bonf_p), ]
sig_age_tbl$Raw_p  <- signif(sig_age_tbl$Raw_p, 4)
sig_age_tbl$Bonf_p <- signif(sig_age_tbl$Bonf_p, 4)

# --- Render DT tables (ranked smallest → largest Bonferroni p-value) ---
datatable(
  sig_gender_tbl,
  caption = "Significant probes after Bonferroni correction — Gender comparison (ranked b
  rownames = FALSE,
  options = list(paging = FALSE, dom = 't')
)
```

Probe_ID	Raw_p	Bonf_p
41214_at	3.498e-17	1.154e-15
34477_at	1.173e-10	3.87e-9

Probe_ID	Raw_p	Bonf_p
38446_at	1.466e-9	4.838e-8
35885_at	4.843e-9	1.598e-7
38355_at	8.558e-9	2.824e-7
37583_at	1.093e-8	3.605e-7
40097_at	4.061e-8	0.00000134
36321_at	6.369e-7	0.00002102
36367_at	0.00007329	0.002419
41138_at	0.001175	0.03877
35465_at	0.00143	0.04718

Significant probes after Bonferroni correction – Gender comparison (ranked by adjusted p-value)

▼ Code

```
datatable(
  sig_age_tbl,
  caption = "Significant probes after Bonferroni correction – Age comparison (ranked by a
  rownames = FALSE,
  options = list(paging = FALSE, dom = 't')
)
```

Probe_ID	Raw_p	Bonf_p
36681_at	0.000008052	0.0002416
38236_at	0.0000112	0.0003361
41720_r_at	0.00001194	0.0003583
35277_at	0.00001446	0.0004338
275_at	0.00001543	0.0004629
1217_g_at	0.00001576	0.0004728
33508_at	0.0000201	0.0006029
31608_g_at	0.00002324	0.0006973

Probe_ID	Raw_p	Bonf_p
37053_at	0.00002362	0.0007087
36897_at	0.00002402	0.0007206
33318_at	0.00002993	0.0008979
235_at	0.00003101	0.0009302
36083_at	0.00003314	0.0009941
622_at	0.00003639	0.001092
41719_i_at	0.00003957	0.001187
39647_s_at	0.00004043	0.001213
38055_at	0.00004248	0.001274
33899_at	0.00005165	0.00155
37405_at	0.00005475	0.001643
35422_at	0.00006329	0.001899
183_at	0.00006586	0.001976
41318_g_at	0.0000734	0.002202
41144_g_at	0.00007431	0.002229
36876_at	0.00008613	0.002584
39760_at	0.00009338	0.002801

Significant probes after Bonferroni correction – Age comparison (ranked by adjusted p-value)

Q7) Read in the log2 normalized fragments per kb per million mapped reads (FPKM) data matrix and annotation files. This is RNA-sequencing data that has normalized read counts on a similar scale to microarray intensities.

▼ Code

```
## Q7) Load TCGA BRCA log2(FPKM) and align to SAM (patient-level)

# 1) Read files
tcga_expr <- read.table("tcga_brca_fpkm.txt",
                        header = TRUE, row.names = 1, sep = "\t", check.names = FALSE)
tcga_sam  <- read.table("tcga_brca_fpkm_sam.txt",
                        header = TRUE, sep = "\t", check.names = FALSE)

# 2) Standardize barcodes
expr_ids_std <- gsub("\\.", "-", colnames(tcga_expr))
expr_pat12   <- substr(expr_ids_std, 1, 12)                    # patient barcode (e.g., T

sam_ids_std  <- gsub("\\.", "-", as.character(tcga_sam$bcr_patient_barcode))
sam_pat12    <- substr(sam_ids_std, 1, 12)                     # should already be 12-cha

# 3) Collapse expression to one column per patient by averaging aliquots (if any)
#    rows = genes, columns = samples -> transpose, group rows by patient, then transpose
expr_sum  <- rowsum(t(tcga_expr), group = expr_pat12)          # sum across aliquots per
expr_n    <- as.vector(table(expr_pat12))                      # counts per patient (sort
tcga_expr_pat <- t(expr_sum / expr_n[col(expr_sum)[,1]])       # mean per patient

# Ensure column order follows the rowsum rownames (patient IDs)
stopifnot(identical(colnames(tcga_expr_pat), rownames(expr_sum)))

# 4) Align to SAM order and drop unmatched safely
match_idx <- match(sam_pat12, colnames(tcga_expr_pat))
keep <- !is.na(match_idx)

tcga_expr_pat <- tcga_expr_pat[, match_idx[keep], drop = FALSE]
tcga_sam_aln  <- tcga_sam[keep, , drop = FALSE]

# Rename columns to the patient barcode for clarity
colnames(tcga_expr_pat) <- sam_pat12[keep]

# 5) Create survival-ready fields for downstream steps
tcga_sam_aln$time   <- suppressWarnings(as.numeric(tcga_sam_aln$months_to_event))
tcga_sam_aln$status <- ifelse(toupper(tcga_sam_aln$vital_status) == "DECEASED", 1L, 0L)
```

Q8) Use grep to subset the data matrix only by gene ‘GATA3’ and make sure to cast this vector to numeric.

▼ Code

```
# Use the patient-level matrix from Q7
expr_mat <- tcga_expr_pat  # genes × patients

# 1) Locate GATA3 (case-insensitive, supports ENSG/pipe suffixes)
```

```
gata_rows <- grep("^GATA3(\\b|\\|)", rownames(expr_mat), ignore.case = TRUE)
stopifnot(length(gata_rows) >= 1)

# 2) Extract and cast to numeric
gata_mat <- expr_mat[gata_rows, , drop = FALSE]
gata3 <- if (nrow(gata_mat) == 1) as.numeric(gata_mat[1, ]) else as.numeric(colMeans(gata
names(gata3) <- colnames(expr_mat)

# 3) Create a tidy dataframe for display
gata_tbl <- data.frame(
  Sample_ID = names(gata3),
  GATA3_log2FPKM = signif(gata3, 4),
  stringsAsFactors = FALSE
)

# 4) Render interactive table
datatable(
  gata_tbl,
  caption = "GATA3 expression across TCGA BRCA samples (log₂ FPKM)",
  rownames = FALSE,
  options = list(pageLength = 10)
)
```

Show 10 entries Search:

Sample_ID	GATA3_log2FPKM
TCGA-FD-A3NA	14.02
TCGA-FD-A3N6	4.386
TCGA-DK-A2I4	11.23
TCGA-DK-A3IK	13.59
TCGA-BT-A20V	10.27
TCGA-BT-A20O	8.096
TCGA-BT-A20P	14.12
TCGA-BT-A20U	10.45
TCGA-CF-A1HR	12.27
TCGA-CF-A1HS	9.092

GATA3 expression across TCGA BRCA samples (log_2 FPKM)

Showing 1 to 10 of 119 entries Previous 1 2 3 4 5 ... 12 Next

Q9) Create a binary (1/0) vector for the patients where the upper 25% expression of GATA3 is coded as 1 and all other patients are coded as 0. Call this new variable 'group'.

▼ Code

```
## Q9) Create binary vector for upper 25% GATA3 expression

# 1) Ensure GATA3 vector is numeric and aligned to patient order
stopifnot(is.numeric(gata3))

# 2) Calculate the 75th percentile cutoff
gata_cut <- quantile(gata3, 0.75, na.rm = TRUE)

# 3) Define binary group: 1 = upper 25% (high expression), 0 = remaining 75%
group <- ifelse(gata3 >= gata_cut, 1L, 0L)
names(group) <- names(gata3)

# 4) Combine with expression for clarity
gata_group_tbl <- data.frame(
  Sample_ID = names(gata3),
  GATA3_log2FPKM = signif(gata3, 4),
  Group = group,
  stringsAsFactors = FALSE
)

# 5) Visualize
datatable(
  gata_group_tbl,
  caption = "Binary grouping of patients by GATA3 expression (upper 25% = 1)",
  rownames = FALSE,
)
```

Show 10 entries Search:

Sample_ID	GATA3_log2FPKM	Group
TCGA-FD-A3NA	14.02	1
TCGA-FD-A3N6	4.386	0
TCGA-DK-A2I4	11.23	0
TCGA-DK-A3IK	13.59	1

Sample_ID	GATA3_log2FPKM	Group
TCGA-BT-A20V	10.27	0
TCGA-BT-A20O	8.096	0
TCGA-BT-A20P	14.12	1
TCGA-BT-A20U	10.45	0
TCGA-CF-A1HR	12.27	0
TCGA-CF-A1HS	9.092	0

Binary grouping of patients by GATA3 expression (upper 25% = 1)

Showing 1 to 10 of 119 entries Previous 1 2 3 4 5 ... 12 Next

Q10) Create a data matrix with the 'group' variable you created in #9 and the remaining variables in the annotation file.

▼ Code

```
## Q10) Merge binary GATA3 group with clinical annotation

# 1) Make sure identifiers align between GATA3 vector and SAM
expr_ids <- substr(names(gata3), 1, 12)
sam_ids  <- substr(gsub("\\.", "-", tcga_sam_aln$bcr_patient_barcode), 1, 12)

# 2) Match annotation rows to GATA3 order
match_idx <- match(expr_ids, sam_ids)

# 3) Combine into one data frame
gata_clin_df <- data.frame(
  Patient_ID = expr_ids,
  GATA3_log2FPKM = signif(gata3, 4),
  Group = group,
  tcga_sam_aln[match_idx, c(
    "age_at_initial_pathologic_diagnosis",
    "gender",
    "vital_status",
    "months_to_event",
    "time",
    "status"
  )],
  stringsAsFactors = FALSE
)
```

```
# 4) Display as a DT table
datatable(
  gata_clin_df,
  caption = "Merged TCGA BRCA annotation with GATA3 grouping (Group = 1 → high expression
  rownames = FALSE,
)
```

Show 10 entries Search:

Patient_ID	GATA3_log2FPKM	Group	age_at_initial_pathologic_diagnosis	gend
TCGA-FD-A3NA	14.02	1	60	MALE
TCGA-FD-A3N6	4.386	0	43	FEMA
TCGA-DK-A2I4	11.23	0	79	MALE
TCGA-DK-A3IK	13.59	1	87	MALE
TCGA-BT-A20V	10.27	0	59	FEMA
TCGA-BT-A20O	8.096	0	75	MALE
TCGA-BT-A20P	14.12	1	81	FEMA
TCGA-BT-A20U	10.45	0	70	FEMA
TCGA-CF-A1HR	12.27	0	62	MALE
TCGA-CF-A1HS	9.092	0	75	FEMA

Merged TCGA BRCA annotation with GATA3 groupir

Showing 1 to 10 of 119 entries Previous 1 2 3 4 5 ... 12 Next

Q11) Run a Kaplan-Meier (KM) analysis to determine if a difference in survival experience exists between the two GATA3 expression groups using the survdiff function. Extract the p-value from the chi squared test output.

▼ Code

```
## Q11) Kaplan–Meier (KM) survival comparison: high vs. low GATA3 expression

library(survival)
```

```
Warning: package 'survival' was built under R version 4.3.3
```

▼ Code

```
# 1) Prepare survival object
# Ensure time and status are numeric and aligned
surv_obj <- Surv(time = gata_clin_df$time, event = gata_clin_df$status)

# 2) Run log-rank test between groups (1 = high GATA3, 0 = low)
km_test <- survdiff(surv_obj ~ gata_clin_df$Group)

# 3) Extract p-value from chi-squared statistic
chisq_val <- km_test$chisq
p_val <- 1 - pchisq(chisq_val, df = 1)

# 4) Display results
p_val_fmt <- signif(p_val, 4)
cat("Log-rank test Chi-squared =", round(chisq_val, 3),
    "with p-value =", p_val_fmt, "\n")
```

```
Log-rank test Chi-squared = 3.215 with p-value = 0.07296
```

Result interpretation: Although the GATA3-high group showed a modest survival advantage relative to the GATA3-low group, the difference did not reach statistical significance ($\chi^2 = 3.215$, $p = 0.073$). This suggests a possible but inconclusive association between GATA3 expression and patient survival in the TCGA BRCA cohort.

Q12) Now run a Cox proportion hazard (PH) regression model on just the grouping variable (i.e. no other covariates) and extract both the p-value and hazard ratio from the output.

▼ Code

```
## Q12) Cox Proportional Hazards (PH) model using GATA3 group

# 1) Create survival object
surv_obj <- Surv(time = gata_clin_df$time, event = gata_clin_df$status)

# 2) Fit Cox PH model with GATA3 group as the only predictor
cox_model <- coxph(surv_obj ~ gata_clin_df$Group)

# 3) Summarize results
cox_summary <- summary(cox_model)

# 4) Extract hazard ratio (HR) and p-value
hr <- signif(cox_summary$coef[,"exp(coef)"], 4)
p_val <- signif(cox_summary$coef[,"Pr(>|z|)"], 4)

cat("Hazard Ratio (HR) =", hr, "\n")
```

```
Hazard Ratio (HR) = 0.4266
```

▼ Code

```
cat("p-value =", p_val, "\n")
```

```
p-value = 0.08162
```

Result Interpretation: The Cox PH model (HR = 0.43, p = 0.082) indicates that higher GATA3 expression is associated with a lower hazard of death, though this relationship does not reach statistical significance. This aligns with the Kaplan–Meier result showing a similar trend toward improved survival among patients with elevated GATA3 expression.

Q13) Run the survfit() function only on the grouping variable (i.e. no other covariates) and plot the KM curves, being sure to label the two groups with a legend, two different colors for each line, and provide the KM p-value, Cox PH p-value, Cox PH hazard ratio, and sample sizes all in each of the two groups all on the plot.

▼ Code

```
## Q13) KM curves (survfit) for GATA3 groups with on-plot stats

library(survival)

# ----- data prep (complete cases only) -----
dat <- gata_clin_df
cc  <- complete.cases(dat$time, dat$status, dat$Group)
dat <- dat[cc, ]
```

```r
# n per group
n0 <- sum(dat$Group == 0, na.rm = TRUE)  # low (bottom 75%)
n1 <- sum(dat$Group == 1, na.rm = TRUE)  # high (top 25%)

# ----- KM fit (no covariates) -----
fit <- survfit(Surv(time, status) ~ Group, data = dat)

# ----- KM (log-rank) p-value -----
km_test <- survdiff(Surv(time, status) ~ Group, data = dat)
km_p <- 1 - pchisq(km_test$chisq, df = 1)

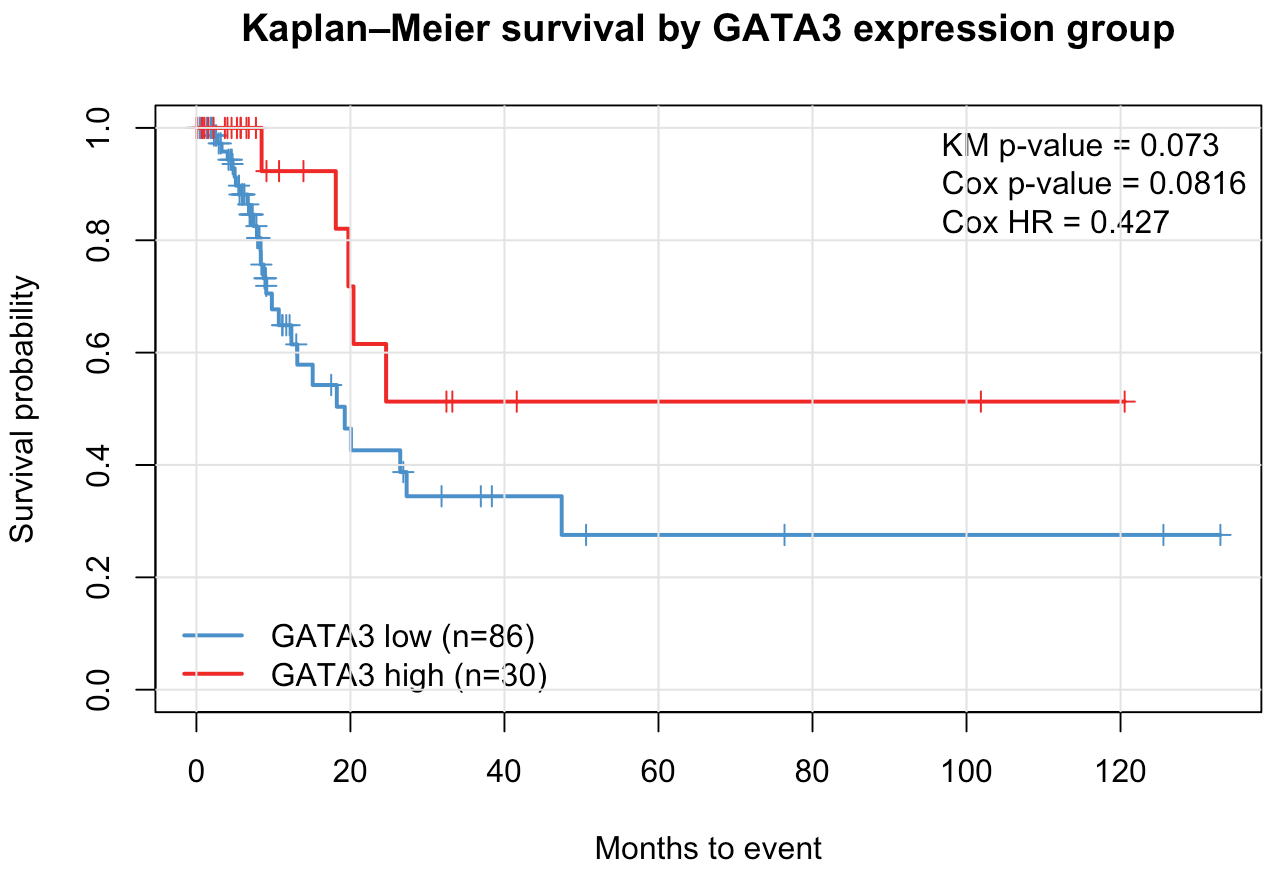
# ----- Cox PH (univariate) -----
cox_mod <- coxph(Surv(time, status) ~ Group, data = dat)
cx      <- summary(cox_mod)
cox_hr  <- cx$coef[,"exp(coef)"]
cox_p   <- cx$coef[,"Pr(>|z|)"]

# ----- plot -----
cols <- c("steelblue3", "firebrick2")   # 0 = blue (low), 1 = red (high)
plot(fit,
     col = cols, lwd = 2, mark.time = TRUE, conf.int = FALSE,
     xlab = "Months to event", ylab = "Survival probability",
     main = "Kaplan–Meier survival by GATA3 expression group")

legend("bottomleft",
       legend = c(paste0("GATA3 low (n=", n0, ")"),
                  paste0("GATA3 high (n=", n1, ")")),
       col = cols, lwd = 2, bty = "n")

# stats block shown on the figure
stats_text <- c(
  paste0("KM p-value = ", signif(km_p, 3)),
  paste0("Cox p-value = ", signif(cox_p, 3)),
  paste0("Cox HR = ", signif(cox_hr, 3))
)
legend("topright", legend = stats_text, bty = "n")

# (Optional) add light grid
abline(h = seq(0, 1, by = 0.2), v = pretty(dat$time), col = "gray90", lwd = 1)
```



Q14) Does this result agree with the Mehra et al, study result?

The present TCGA analysis shows that patients in the top quartile of GATA3 expression exhibit a trend toward improved survival (HR = 0.43, p = 0.082), indicating a lower hazard of death compared with those with lower expression. This agrees directionally with the results of Mehra et al. (2005), who found that low GATA3 expression was associated with higher tumor grade, ER-negativity, and markedly poorer disease-free and overall survival (univariate HR = 0.31, p = 0.009). The difference lies in statistical strength—our analysis did not reach significance, likely due to smaller effective sample size and lack of adjustment for confounders such as tumor subtype or ER status. Overall, both datasets support GATA3 as a favorable prognostic marker in breast cancer, validating the biological conclusion of Mehra et al. while illustrating the reduced sensitivity of a single-variable analysis on normalized RNA-seq data.